The rectangular coordinate system in space

A third axis, and the two formulas — distance and midpoint — that everything else in the chapter is built on.

LESSON 1–3 | 3 × 40 MIN | GEOMETRY 11, §1 | P1 · 3-D COORDINATES

LESSON CLOCK

14' 22' 30' 24' 24' 21'

The third axis	14 min
Plotting and reading points	22 min
The distance formula	30 min
The midpoint formula	24 min
The sphere	24 min
Practice and homework	21 min

LEARNING OBJECTIVES

- Plot a point from its three coordinates and read coordinates from a diagram.
- Use the distance formula in space.
- Use the midpoint formula in space.
- Find the equation of a sphere from its centre and radius.

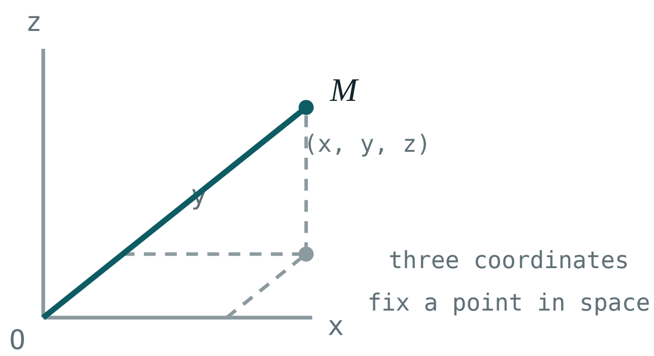
TEXTBOOK

National	pp. 3–16
Cambridge	Core & Extended, pp. 318–324
Workbook	Exercise 1.1

01 Explanation

A third axis

Add an axis Oz perpendicular to both Ox and Oy. Every point of space then has exactly three coordinates (x, y, z) — how far along, how far across, how far up.



The point (3, 4, 5) and the box that locates it. Read along, across, then up.

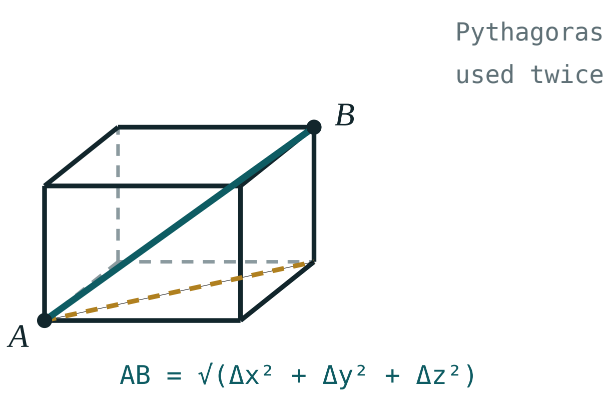
OBJECT	DESCRIPTION
the three axes	Ox, Oy, Oz
the three coordinate planes	$z = 0, y = 0, x = 0$
octants	8
a point on the x-axis	$(a, 0, 0)$
a point in the xy-plane	$(a, b, 0)$

RIGHT-HANDED CONVENTION

Point the fingers of the right hand along Ox and curl them towards Oy; the thumb points along Oz. Every formula in this chapter assumes it.

Distance

Pythagoras, used twice — once in the base plane and once vertically:



The base diagonal first, then the vertical rise. One right triangle inside another.

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

From the origin it shortens to $OA = \sqrt{x^2 + y^2 + z^2}$. For $(3, 4, 12)$ that is $\sqrt{9 + 16 + 144} = 13$.

SQUARE THE DIFFERENCES, NOT THE COORDINATES

$A(1, 2, 3)$ and $B(4, 6, 15)$ give $\sqrt{9 + 16 + 144} = 13$ — from the **differences** $3, 4, 12$. Squaring the coordinates themselves is the commonest slip.

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$$

Each coordinate is averaged independently — the plane formula with one more entry.

More generally, the point dividing AB in the ratio $m : n$ from A is

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n}, \frac{nz_1 + mz_2}{m + n}\right)$$

and the midpoint is the case $m = n$.

The sphere

A sphere is the set of points a fixed distance from a centre. Writing that with the distance formula and squaring gives its equation directly:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

Centre $(2, -1, 3)$, radius 5: $(x - 2)^2 + (y + 1)^2 + (z - 3)^2 = 25$.

GOING BACKWARDS

Given $x^2 + y^2 + z^2 - 4x + 6y - 12 = 0$, complete the square in each variable: $(x - 2)^2 + (y + 3)^2 + z^2 = 25$. Centre $(2, -3, 0)$, radius 5.

02 Worked examples

EXAMPLE 1 **Find the distance between $A(1, -2, 3)$ and $B(4, 2, 15)$.**

1. Differences $3, 4, 12$.
2. $9 + 16 + 144 = 169$
3. $\sqrt{169} = 13$

ANSWER

13

EXAMPLE 2 **$A(2, -3, 5)$ and $M(4, 1, 0)$ is the midpoint of AB. Find B.**

1. $\dfrac{2 + x}{2} = 4 \Rightarrow x = 6$
2. $\dfrac{-3 + y}{2} = 1 \Rightarrow y = 5$
3. $\dfrac{5 + z}{2} = 0 \Rightarrow z = -5$

ANSWER

$B(6, 5, -5)$

EXAMPLE 3 **Find the centre and radius of $x^2 + y^2 + z^2 - 6x + 4z - 3 = 0$.**

1. $(x - 3)^2 - 9$
2. $(z + 2)^2 - 4$
3. $(x - 3)^2 + y^2 + (z + 2)^2 = 16$
4. Read off.

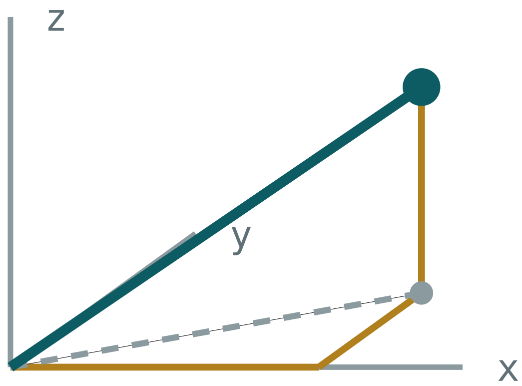
ANSWER

centre $(3, 0, -2)$, radius 4

03 Interactive model

Move the point and watch each coordinate and the distance from the origin update.

A point in space



x **3** | y **2** | z **2** | OM **4.12** | base diagonal **3.61**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Coordinate system in space	Fazoviy koordinatalar sistemasi	Пространственная система координат
Ordinate / abscissa / applicate	Ordinata / abssissa / applikata	Ордината / абсцисса / аппликата
Octant	Oktant	Октант
Coordinate plane	Koordinata tekisligi	Координатная плоскость
Distance between two points	Ikki nuqta orasidagi masofa	Расстояние между точками
Midpoint of a segment	Kesma o‘rtasi	Середина отрезка
Sphere	Sfera	Сфера
Projection onto a plane	Tekislikka proyeksiya	Проекция на плоскость
Origin	Koordinatalar boshi	Начало координат

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The distance from the origin to $(2, 3, 6)$ is:

- 11
- 7
- 9
- $\sqrt{11}$

The midpoint of $(0,0,0)$ and $(4, 6, 8)$ is:

- $(2, 3, 4)$
- $(4, 6, 8)$
- $(1, 2, 3)$
- $(8, 12, 16)$

A point in the xy-plane has:

- $x = 0$
- $y = 0$
- $z = 0$
- all zero

How many octants?

- 4
- 6
- 8
- 12

$(x - 1)^2 + y^2 + (z + 2)^2 = 9$ has radius:

- 9
- 3
- $\sqrt{3}$
- 81

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Distance from O to $(1, 2, 2)$

   ANSWER 3

2. Distance from O to $(2, 3, 6)$

   ANSWER 7

3. Distance from O to $(3, 4, 12)$

   ANSWER 13

4. Midpoint of $(0,0,0)$ and $(2, 4, 6)$

   ANSWER $(1, 2, 3)$

5. Midpoint of $(1,1,1)$ and $(3,5,7)$

   ANSWER $(2, 3, 4)$

6. Radius of $x^2 + y^2 + z^2 = 36$

   ANSWER 6

7. A point on the z-axis has form

   ANSWER $(0, 0, c)$

Medium · the standard question

7 PROBLEMS

1. Distance $A(1,2,3)$ to $B(4,6,15)$

   ANSWER 13

2. Distance $A(-1,0,2)$ to $B(3,4,2)$

   ANSWER $4\sqrt{2}$

3. Midpoint of $(-2, 5, 1)$ and $(6, -1, 7)$

   ANSWER $(2, 2, 4)$

4. Sphere, centre $(1,2,3)$ radius 4 — its equation

   ANSWER $(x-1)^2 + (y-2)^2 + (z-3)^2 = 16$

5. Centre and radius of $x^2 + y^2 + z^2 - 4x = 0$

   ANSWER $(2,0,0)$, 2

6. $M(3,1,2)$ is the midpoint of $A(1,-1,0)B$. Find B

   ANSWER $(5, 3, 4)$

7. Is $(1,2,2)$ on the sphere $x^2+y^2+z^2 = 9$?

   ANSWER yes

Hard · stretch and reason

7 PROBLEMS

1. Show $A(1,2,3)$, $B(3,4,5)$, $C(5,6,7)$ are collinear

   ANSWER Equal steps of $(2,2,2)$

2. Find z so $(1, 2, z)$ is 7 from $(4, 6, 1)$

   ANSWER $z = 1 \pm 2\sqrt{6}$

3. Point dividing $A(1,0,2)B(7,6,8)$ in ratio $1:2$

   ANSWER $(3, 2, 4)$

4. Centre and radius of $x^2+y^2+z^2 - 2x + 4y - 6z - 11 = 0$

   ANSWER $(1,-2,3), 5$

5. Find the point equidistant from $(0,0,0)$, $(2,0,0)$, $(0,2,0)$, $(0,0,2)$

   ANSWER $(1, 1, 1)$

6. Show $A(0,0,0)$, $B(4,0,0)$, $C(4,3,0)$, $D(0,3,0)$ form a rectangle

   ANSWER Opposite sides equal, diagonals equal 5

7. A cube has vertices $(0,0,0)$ and (a,a,a). Find the space diagonal

   ANSWER $a\sqrt{3}$

07 Homework

Homework — 6 tasks

Show the differences before squaring in every distance question.

1. Find the distance between $A(2, -1, 4)$ and $B(5, 3, 16)$.
2. Find the midpoint of $A(-3, 2, 7)$ and $B(5, -4, 1)$.
3. $M(1, 0, 2)$ is the midpoint of AB with $A(-2, 3, 5)$. Find B.
4. Write the equation of the sphere with centre $(-1, 2, 0)$ and radius 6.
5. Find the centre and radius of $x^2 + y^2 + z^2 + 2x - 8y + 4z + 5 = 0$.

6. Show that $A(1,1,1)$, $B(3,3,3)$, $C(6,6,6)$ are collinear and find the ratio $AB : BC$.

Home / Grade 11 / Geometry / Quarter 1

Vectors in space and operations on them

The same three operations as in the plane, with one extra component — and the position vector that makes coordinates and vectors the same thing.

LESSON 4–5 | 2 × 40 MIN | GEOMETRY 11, §2 | P1 · VECTORS

LESSON CLOCK

10' | 20' | 22' | 20' | 18' | 10'

What carries over from the plane	10 min
Components and the basis	20 min
The three operations	22 min
Magnitude and unit vectors	20 min
Position vectors	18 min
Homework	10 min

LEARNING OBJECTIVES

- Write a vector in component and in unit-vector form.
- Add, subtract and multiply a vector by a scalar.
- Find the magnitude of a vector and a unit vector in its direction.
- Use position vectors to express AB as b − a.

TEXTBOOK

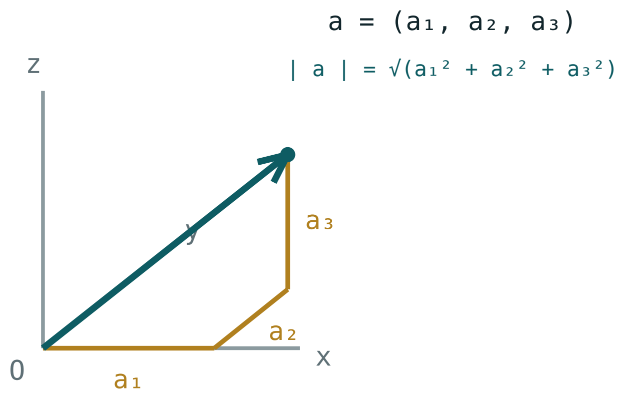
National	pp. 17–28
Cambridge	Core & Extended, pp. 325–334
Workbook	Exercise 2.1

01 Explanation

One extra component, nothing else new

A vector in space is written in components or on the basis i, j, k:

$$a = (a_1, a_2, a_3) = a_1 i + a_2 j + a_3 k$$



A vector as an arrow, and as the three steps along the axes that build it.

> VECTOR OR POINT?
>
> $(3, 4, 5)$ as a **point** is a place. As a **vector** it is a displacement — three along, four across, five up — which may start anywhere. Two vectors are equal when their components agree, wherever they are drawn.

The three operations

OPERATION	RULE	EXAMPLE WITH $A = (1,2,3)$, $B = (4,0,-1)$
addition	$(a_1+b_1, a_2+b_2, a_3+b_3)$	$(5, 2, 2)$
subtraction	$(a_1-b_1, a_2-b_2, a_3-b_3)$	$(-3, 2, 4)$
scalar multiple	(ka_1, ka_2, ka_3)	$3a = (3, 6, 9)$

Geometrically nothing changes: addition is still the triangle rule, and ka is still a stretched by $|k|$, reversed if k is negative.

> VECTORS ARE ADDED COMPONENTWISE
>
> There is no “cross-multiplying” of components in addition. Each component minds its own axis.

Magnitude and unit vectors

$$|a| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

It is the distance formula from the origin, and it obeys the same rules as in the plane: $|ka| = |k| \cdot |a|$, and $|a| = 0$ only for the zero vector.

A **unit vector** has magnitude 1. To build one in the direction of a, divide by the magnitude:

$$\hat{a} = \frac{\mathrm{a}}{|\mathrm{a}|}$$

For $a = (2, -1, 2)$: $|a| = 3$, so $\hat{a} = (\frac{2}{3}, -\frac{1}{3}, \frac{2}{3})$.

DIVIDE THE VECTOR, NOT THE COMPONENTS SEPARATELY IN YOUR HEAD

Every component is divided by the **same** number $|a|$. Check by confirming the new magnitude is 1.

Position vectors

THE BRIDGE BETWEEN POINTS AND VECTORS

The **position vector** of A is OA, written a, and its components are simply the coordinates of A. Then for any two points:

$$AB = b - a$$

“Head minus tail” — the destination’s position vector minus the start’s.

So $A(1, 2, 3)$ to $B(4, 6, 15)$ gives $AB = (3, 4, 12)$, of length 13 — the same answer as the distance formula, because it **is** the distance formula.

The midpoint of AB has position vector $\frac{\mathrm{a} + \mathrm{b}}{2}$, which is the midpoint formula written once instead of three times.

02 Worked examples

EXAMPLE 1 $\boldsymbol{a} = (2, -1, 3), \boldsymbol{b} = (1, 4, -2)$. **Find** $2a - 3b$ **and its magnitude.**

1. $2a = (4, -2, 6)$
2. $3b = (3, 12, -6)$
3. $2a - 3b = (1, -14, 12)$
4. $|\cdot| = \sqrt{1 + 196 + 144} = \sqrt{341} \approx 18.47$

ANSWER

$(1, -14, 12)$, magnitude $\sqrt{341} \approx 18.5$

EXAMPLE 2 **Find a unit vector in the direction of** $\boldsymbol{a} = (6, -2, 3)$.

1. $|a| = \sqrt{36 + 4 + 9} = 7$
2. $\hat{a} = \frac{1}{7}(6, -2, 3)$
3. Check: $\frac{36 + 4 + 9}{49} = 1$ ✓

ANSWER

$\left(\frac{6}{7}, -\frac{2}{7}, \frac{3}{7}\right)$

EXAMPLE 3 **$A(3, 0, -1)$, $B(1, 4, 5)$. Find AB, its length, and the midpoint.**

1. $AB = b - a = (-2, 4, 6)$
2. $|AB| = \sqrt{4 + 16 + 36} = \sqrt{56} = 2\sqrt{14}$
3. $M = \dfrac{a + b}{2} = (2, 2, 2)$

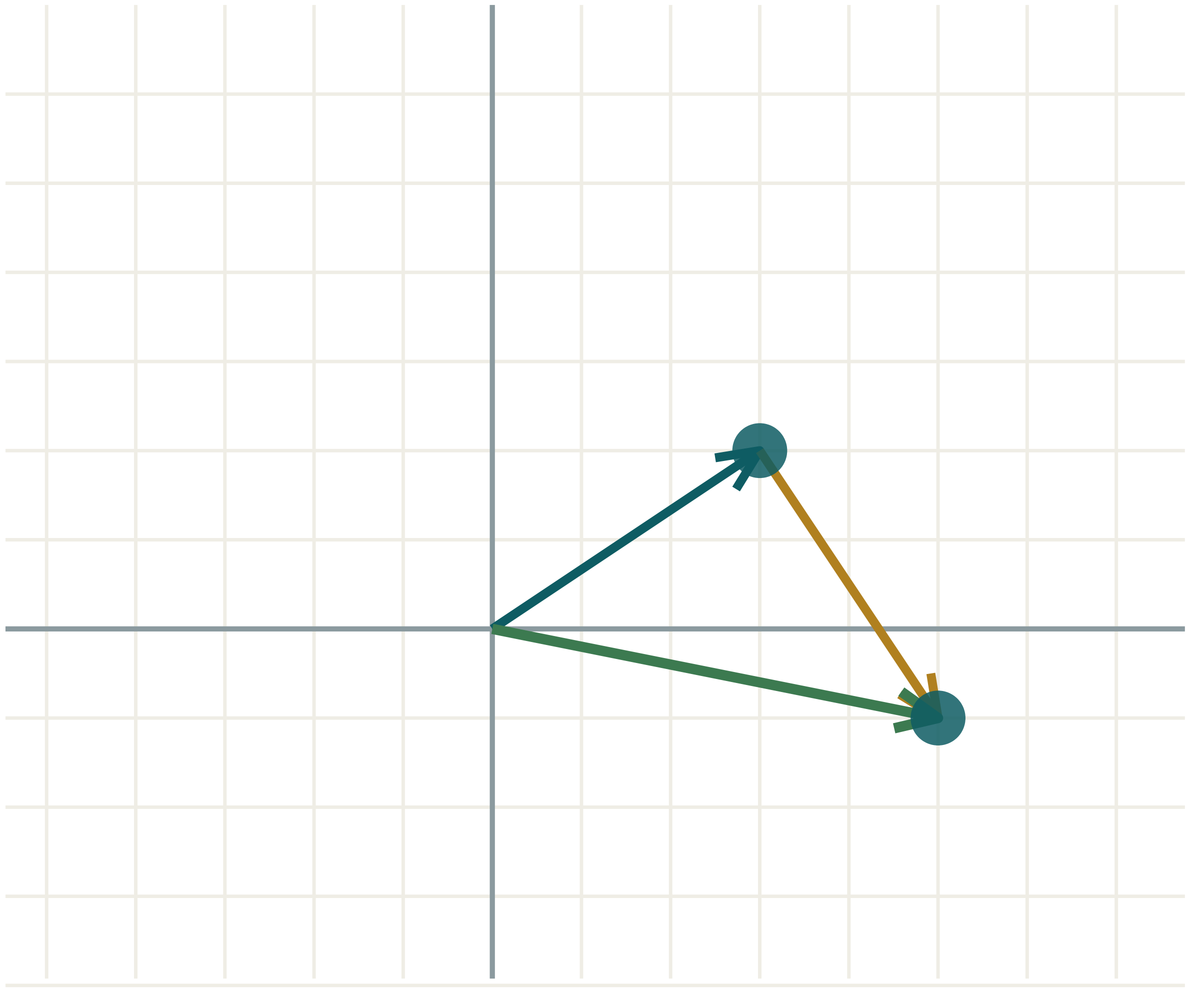
ANSWER

$AB = (-2, 4, 6),\ 2\sqrt{14},\ M(2, 2, 2)$

03 Interactive model

Change the components and watch the arrow and its length move together.

Vector arithmetic



a (3; 2) | b (2; -3) | a + b (5; -1) | |a| 3.61 | |b| 3.61 | |a + b| 5.1

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Vector	Vektor	Вектор
Component	Komponenta	Компонента
Position vector	Radius-vektor	Радиус-вектор
Magnitude (modulus)	Uzunlik (modul)	Длина (модуль)
Unit vector	Birlik vektor	Единичный вектор
Zero vector	Nol vektor	Нулевой вектор
Scalar multiplication	Skalyarga ko‘paytirish	Умножение на скаляр
Equal vectors	Teng vektorlar	Равные векторы
Basis vectors i, j, k	Bazis vektorlar	Базисные векторы

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$|(3, 4, 12)|$ is:

- 19
- 13
- 12
- $\sqrt{19}$

AB in position vectors is:

- $a + b$
- $b - a$
- $a - b$
- $\frac{a + b}{2}$

A unit vector has magnitude:

- 0
- 1
- $|a|$
- any

$(1,2,3) + (4,0,-1)$ is:

- $(5,2,2)$
- $(4,0,-3)$
- $(5,2,4)$
- $(-3,2,4)$

$|3a|$ equals:

- $|a|$
- $3|a|$
- $9|a|$
- $|a|^3$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $(1,2,3) + (3,2,1)$

   ANSWER $(4,4,4)$

2. $(5,0,2) - (1,3,2)$

   ANSWER $(4,-3,0)$

3. $2(1,-2,3)$

   ANSWER $(2,-4,6)$

4. $|(1,2,2)|$

   ANSWER 3

5. $|(0,3,4)|$

   ANSWER 5

6. $|(2,3,6)|$

   ANSWER 7

7. Unit vector along $(0,0,5)$

   ANSWER $(0,0,1)$

Medium · the standard question

7 PROBLEMS

1. $a = (2,-1,3)$, $b = (1,0,4)$; find $a + 2b$

   ANSWER $(4,-1,11)$

2. Same; find $3a - b$

   ANSWER $(5,-3,5)$

3. $|(6,-2,3)|$

   ANSWER 7

4. Unit vector along $(6,-2,3)$

   ANSWER $\frac{1}{7}(6,-2,3)$

5. $A(1,2,3)$, $B(4,6,15)$; find AB

   ANSWER $(3,4,12)$

6. Same; find $|AB|$

   ANSWER 13

7. Midpoint of $A(1,2,3)$ and $B(5,6,7)$

   ANSWER $(3,4,5)$

Hard · stretch and reason 7 PROBLEMS

1. Find k so $|(k, 2, 2)| = 3$

ANSWER $k = \pm 1$

2. Find a unit vector opposite to $(1, 2, -2)$

ANSWER $-\frac{1}{3}(1,2,-2)$

3. $a = (1,2,2)$, $b = (2,-1,2)$; find $|a + b|$

ANSWER $\sqrt{34} \approx 5.83$

4. Show $|a + b| \le |a| + |b|$ for those vectors

ANSWER $5.83 \le 6$

5. Find t so $(1, t, 2)$ is parallel to $(3, 6, 6)$

ANSWER $t = 2$

6. Express $(5, 4, 3)$ in terms of i, j, k

ANSWER $5i + 4j + 3k$

7. $A(2,1,0)$, $B(4,5,6)$; find the point P on AB with $AP : PB = 1 : 3$

ANSWER $(2.5, 2, 1.5)$

07 Homework

Homework — 5 tasks

Check every unit vector by confirming its magnitude is 1.

1. $a = (3, -1, 2)$, $b = (0, 4, -2)$. Find $a + b$, $a - b$ and $2a + 3b$.
2. Find $|a|$ and a unit vector along a for $a = (2, 3, 6)$.
3. $A(0, 1, -2)$, $B(6, 5, 4)$. Find AB, $|AB|$ and the midpoint.
4. Find k so that $|(k, 4, 3)| = 13$.

5. Explain in three sentences the difference between the point $(2,3,4)$ and the vector $(2,3,4)$.

Collinear and coplanar vectors

Two tests, both from the definition: one scalar for collinearity, two for coplanarity.

LESSON 6 | 1 × 40 MIN | GEOMETRY 11, §3 | P1 · VECTORS

LESSON CLOCK

8' | 12' | 14' | 8' | 3'

Collinearity	8 min
Proving three points collinear	12 min
Coplanarity	14 min
Practice	8 min
Homework	3 min

LEARNING OBJECTIVES

- Test two vectors for parallelism.
- Prove three points are collinear using vectors.
- Express a vector as a linear combination of two others.
- Decide whether three vectors are coplanar.

TEXTBOOK

National	pp. 29–34
Cambridge	Core & Extended, pp. 335–338
Workbook	Exercise 3.1

01 Explanation

Collinear vectors

THE TEST

a and b are **collinear** (parallel) exactly when $b = ka$ for some scalar k — equivalently, when their components are in a constant ratio: $\frac{b_1}{a_1} = \frac{b_2}{a_2} = \frac{b_3}{a_3}$.

$(2, -4, 6)$ and $(1, -2, 3)$ are collinear with $k = 2$. $(1, 2, 3)$ and $(2, 4, 5)$ are not: the first two ratios give 2 but the third gives $\frac{5}{3}$.

ALL THREE RATIOS, NOT TWO

Two agreeing ratios prove nothing. This is exactly how a “parallel” answer goes wrong.

Three points collinear

Points A, B, C lie on one line exactly when AB and AC are collinear vectors — they already share the point A, so a common direction is enough.

$$A(1,2,3),\ B(3,5,7),\ C(7,11,15):\ AB = (2,3,4),\ AC = (6,9,12) = 3AB$$

So the three are collinear, and $AB : BC = 1 : 2$.

Coplanar vectors

THE TEST

Three vectors are **coplanar** when one is a linear combination of the other two: $c = \alpha a + \beta b$. Geometrically, all three can be drawn in one plane.

Method. Write the equation componentwise, giving three equations in α and β. Solve two of them and check the third. If it holds, they are coplanar; if it fails, they are not.

$$c = \alpha a + \beta b \Rightarrow \textit{three equations, two unknowns} \Rightarrow \textit{check for consistency}$$

Two vectors are always coplanar — two directions determine a plane. Coplanarity only becomes a real question with three.

02 Worked examples

EXAMPLE 1 **Are $a = (2, -6, 4)$ and $b = (-3, 9, -6)$ collinear?**

1. $\dfrac{-3}{2} = -1.5$

2. $\dfrac{9}{-6} = -1.5$

3. $\dfrac{-6}{4} = -1.5$

   All three agree.

ANSWER

Yes, with $b = -1.5a$

EXAMPLE 2 **Are $A(1,0,2)$, $B(3,2,4)$, $C(6,5,7)$ collinear?**

1. $AB = (2,2,2)$
2. $AC = (5,5,5)$
3. $AC = 2.5\ AB$

ANSWER

Yes

EXAMPLE 3 **Are $a = (1,0,1)$, $b = (0,1,1)$, $c = (2,3,5)$ coplanar?**

1. $c = \alpha a + \beta b$
2. $\alpha = 2$ from the first component.
3. $\beta = 3$ from the second.
4. Third: $\alpha + \beta = 5$ ✓

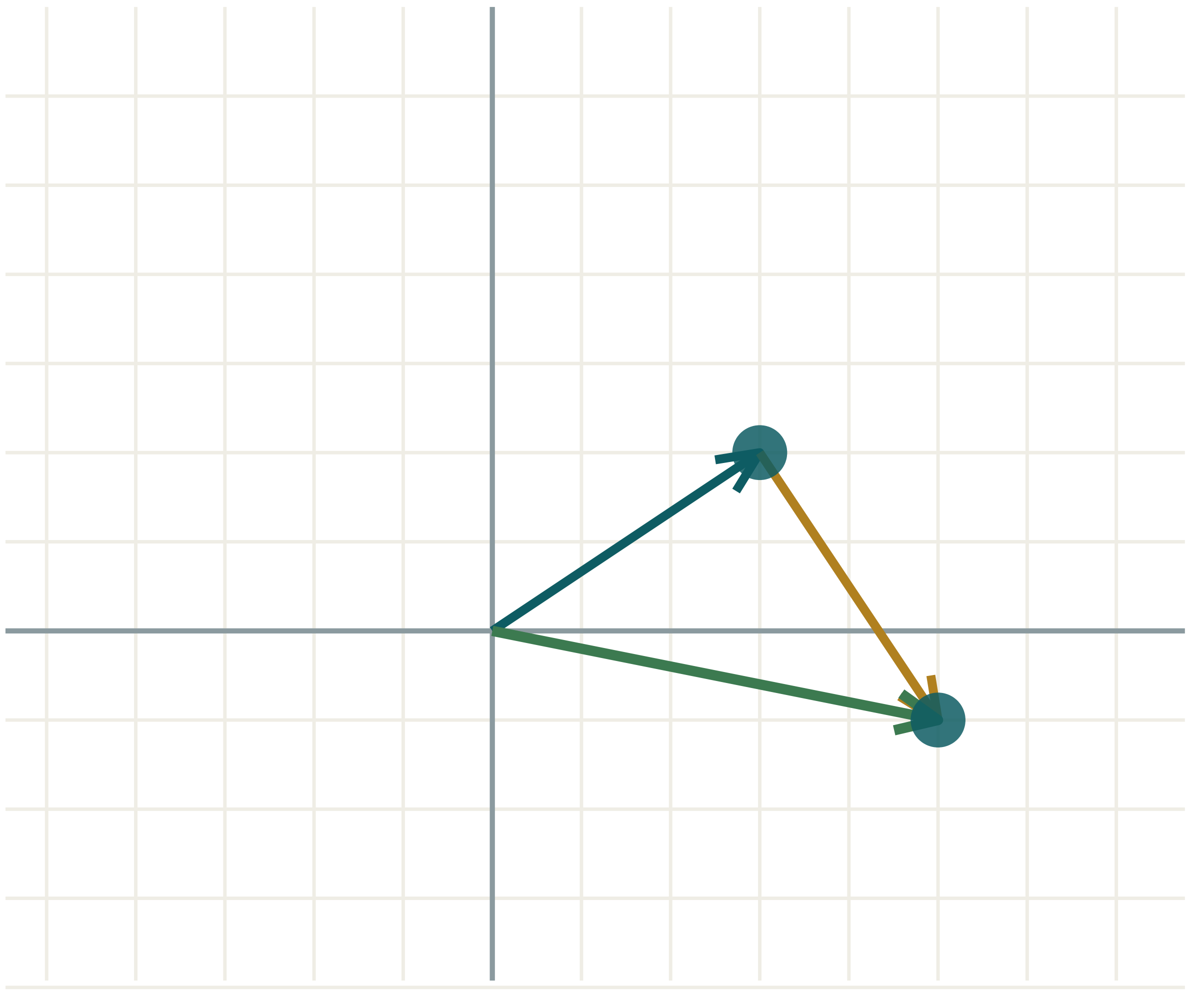
ANSWER

Yes — $c = 2a + 3b$

03 Interactive model

Draw two vectors, then a third, and ask whether a sheet of paper can hold all three.

Combining vectors



a (3; 2) | b (2; -3) | a + b (5; -1) | |a| 3.61 | |b| 3.61 | |a + b| 5.1

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Collinear vectors	Kollinear vektorlar	Коллинеарные векторы
Coplanar vectors	Komplanar vektorlar	Компланарные векторы
Linear combination	Chiziqli kombinatsiya	Линейная комбинация
Parallel vectors	Parallel vektorlar	Параллельные векторы
Decomposition	Yoyish	Разложение
Linearly independent	Chiziqli erkli	Линейно независимые
Scalar coefficient	Skalyar koeffitsient	Скалярный коэффициент
Common point	Umumiy nuqta	Общая точка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

a and b are collinear when:

- $a = b$
- $b = ka$
- $|a| = |b|$
- always

$(1,2,3)$ and $(2,4,6)$ are:

- collinear
- perpendicular
- coplanar only
- unrelated

Three vectors are coplanar when:

- all are parallel
- one is a combination of the others
- all are unit vectors
- never

Two vectors are:

- never coplanar
- always coplanar
- sometimes coplanar
- always collinear

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Are $(1,2,3)$ and $(3,6,9)$ collinear?

   ANSWER yes

2. Are $(1,2,3)$ and $(2,4,7)$ collinear?

   ANSWER no

3. Find k: $(4,8,12) = k(1,2,3)$

   ANSWER $k = 4$

4. Are $(0,0,1)$ and $(0,0,-5)$ collinear?

   ANSWER yes

5. Are two vectors always coplanar?

   ANSWER yes

6. Is $(2,2,2)$ parallel to $(1,1,1)$?

   ANSWER yes

7. $AB = (1,1,1)$, $AC = (3,3,3)$. Are the points collinear?

   ANSWER yes

Medium · the standard question

7 PROBLEMS

1. Are $A(1,0,2)$, $B(3,2,4)$, $C(6,5,7)$ collinear?

   ANSWER yes

2. Are $A(0,0,0)$, $B(1,1,1)$, $C(2,2,3)$ collinear?

   ANSWER no

3. Find t so $(2, t, 6)$ is parallel to $(1, 3, 3)$

   ANSWER $t = 6$

4. Is $(2,3,5)$ in the plane of $(1,0,1)$ and $(0,1,1)$?

   ANSWER yes

5. Express $(3,4,7)$ from $(1,0,1)$ and $(0,1,1)$

   ANSWER $3a + 4b$

6. Are $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ coplanar?

   ANSWER no

7. Find the ratio $AB : BC$ for $A(1,2,3)$, $B(3,5,7)$, $C(7,11,15)$

   ANSWER $1 : 2$

Hard · stretch and reason

7 PROBLEMS

1. Are $(1,2,1)$, $(2,1,3)$, $(4,5,5)$ coplanar?

   ANSWER yes — $c = 2a + b$

2. Are $(1,1,0)$, $(0,1,1)$, $(1,0,2)$ coplanar?

   ANSWER no

3. Find λ so $(1,2,\lambda)$ is coplanar with $(1,0,1)$ and $(0,1,1)$

   ANSWER $\lambda = 3$

4. Prove the midpoints of the sides of a space quadrilateral form a parallelogram

   ANSWER Each midline equals $\frac{1}{2}$ of a diagonal

5. Prove the medians of a triangle are concurrent using vectors

   ANSWER All meet at $\frac{a + b + c}{3}$

6. Show that if $c = \alpha a + \beta b$ then a, b, c cannot span space

   ANSWER They lie in the plane of a and b

7. Find P on AB with $AP : PB = 2:3$ for $A(1,1,1)$, $B(6,11,16)$

   ANSWER $(3, 5, 7)$

07 Homework

Homework — 4 tasks

Check all three ratios, never two.

1. Are $(3, -6, 9)$ and $(-2, 4, -6)$ collinear? Show the ratios.
2. Are $A(2,1,0)$, $B(4,4,3)$, $C(8,10,9)$ collinear? Give $AB : BC$.
3. Show that $(3, 5, 8)$ is coplanar with $(1, 1, 2)$ and $(1, 3, 4)$.
4. Find λ so that $(2, \lambda, 5)$ is coplanar with $(1, 0, 1)$ and $(0, 1, 2)$.

The scalar product, magnitude and the angle between vectors

One multiplication that returns a number instead of a vector — and answers every angle question in space.

LESSON 7–8 | 2 × 40 MIN | GEOMETRY 11, §4 | P1 · SCALAR PRODUCT

LESSON CLOCK

12' | 20' | 20' | 18' | 20' | 10'

Two formulas for one product	12 min
The angle formula	20 min
Perpendicularity	20 min
Projections	18 min
Practice	20 min
Homework	10 min

LEARNING OBJECTIVES

- Compute the scalar product from components and from magnitudes and the angle.
- Find the angle between two vectors.
- Test two vectors for perpendicularity.
- Find the projection of one vector on another.

TEXTBOOK

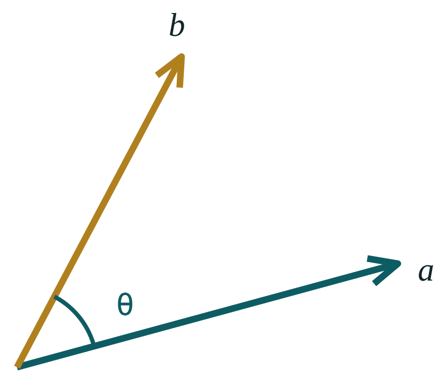
National	pp. 35–46
Cambridge	Core & Extended, pp. 339–348
Workbook	Exercise 4.1

01 Explanation

Two formulas, one number

$a \cdot b = a_1b_1 + a_2b_2 + a_3b_3$ and $a \cdot b = |a||b| \cos \theta$

The first computes it; the second interprets it. Setting them equal is what makes the angle findable.



$a \cdot b = |a|\ |b| \cos\theta = a_1b_1 + a_2b_2 + a_3b_3$

perpendicular exactly when $a \cdot b = 0$

Two vectors and the angle between them, drawn from a common start point.

THE RESULT IS A NUMBER

$a \cdot b$ is a **scalar**, not a vector. Expressions like $(a \cdot b) \cdot c$ as a further dot product are meaningless — you cannot dot a number with a vector.

Properties: $a \cdot b = b \cdot a$, $a \cdot (b + c) = a \cdot b + a \cdot c$, and $a \cdot a = |a|^2$ — which gives a second way to find a magnitude.

The angle

$$\cos\theta = \frac{\mathrm{a} \cdot \mathrm{b}}{|\mathrm{a}||\mathrm{b}|}$$

For $a = (1, 2, 2)$ and $b = (2, -1, 2)$:

$$a \cdot b = 2 - 2 + 4 = 4,\ |a| = 3,\ |b| = 3 \Rightarrow \cos\theta = \frac{4}{9} \Rightarrow \theta \approx 63.6°$$

$A \cdot B$	ANGLE	MEANING
> 0	$\theta < 90°$	the vectors point broadly the same way
$= 0$	$\theta = 90°$	perpendicular
< 0	$\theta > 90°$	broadly opposite

THE SIGN ALONE TELLS YOU A LOT

Before computing an angle, look at the sign of $a \cdot b$. A negative dot product with a final answer of $40°$ is a certain arithmetic error.

Perpendicularity

THE SINGLE MOST USED CONSEQUENCE

$a \perp b \iff a \cdot b = 0$ (for non-zero vectors). No angles, no roots — one line of arithmetic.

$(1, 2, -1)$ and $(3, -1, 1)$: $3 - 2 - 1 = 0$, so they are perpendicular.

This is the test used for every right angle in the rest of the course — the diagonals of a rhombus, the edges of a cuboid, a line perpendicular to a plane.

Projection

The **projection** of a on b is the length of the shadow a casts along the direction of b:

$$proj = |a| \cos\theta = \frac{a \cdot b}{|b|}$$

For $a = (3, 4, 0)$ on $b = (1, 0, 0)$: $\frac{3}{1} = 3$ — the x-component, exactly as expected.

In physics the work done by a force F over a displacement s is $F \cdot s$ — the same idea, and the reason the scalar product was invented.

02 Worked examples

EXAMPLE 1 **Find the angle between $\boldsymbol{a} = (2, 1, -2)$ and $\boldsymbol{b} = (1, 2, 2)$.**

1. $a \cdot b = 2 + 2 - 4 = 0$
2. A zero dot product.
3. No further work is needed.

ANSWER

$90°$ — they are perpendicular

EXAMPLE 2 **Find the angle between $\boldsymbol{a} = (1, 1, 0)$ and $\boldsymbol{b} = (1, 0, 1)$.**

1. $a \cdot b = 1$
2. $|a| = |b| = \sqrt{2}$
3. $\cos\theta = \dfrac{1}{2}$
4. $\theta = 60°$

ANSWER

$60°$

EXAMPLE 3 **Find t so that $(t, 2, 3)$ and $(4, t, -6)$ are perpendicular.**

1. $4t + 2t - 18 = 0$
2. $6t = 18$
3. $t = 3$

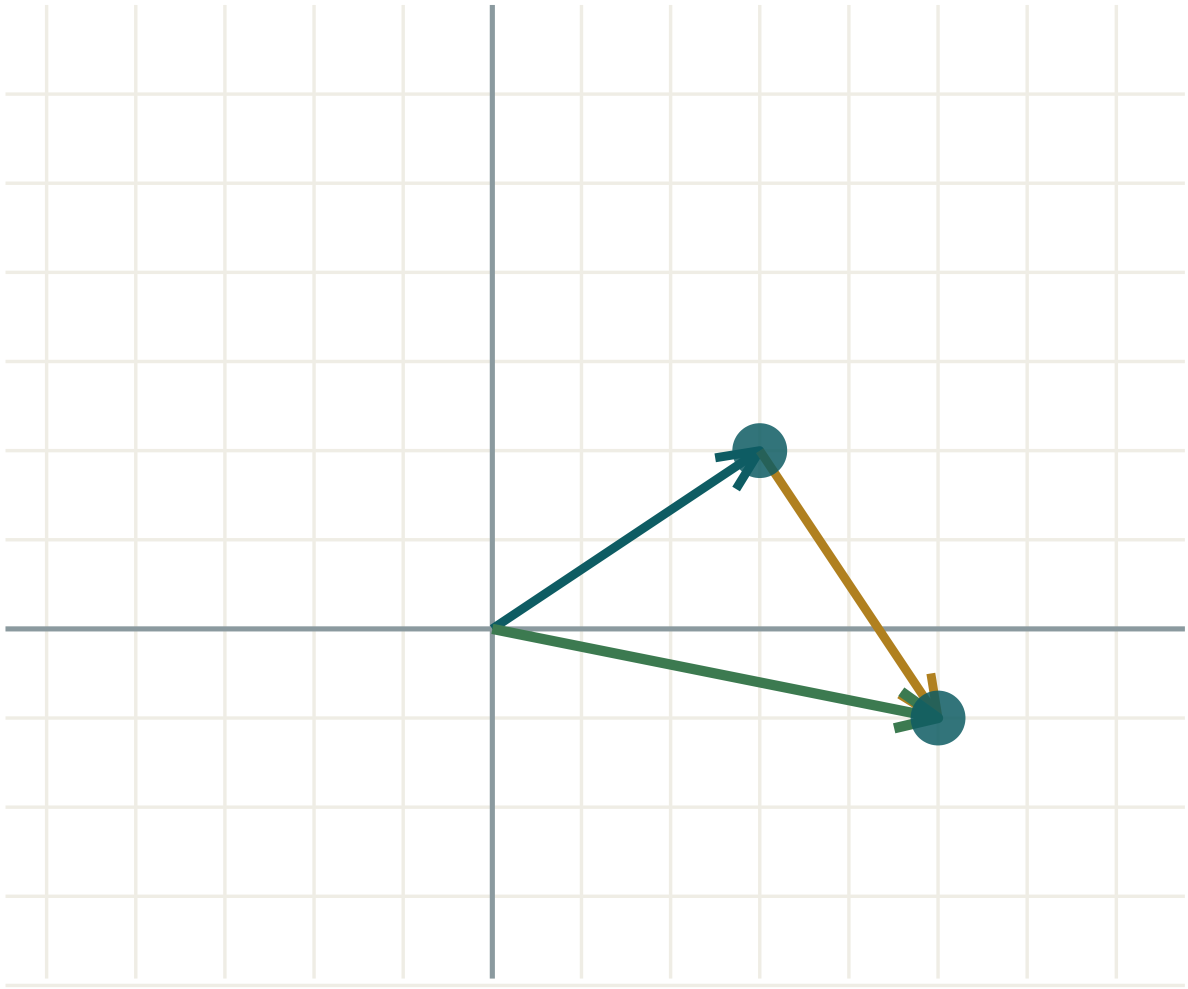
ANSWER

$t = 3$

03 Interactive model

Rotate one vector and watch the dot product pass through zero exactly at the right angle.

The angle between two vectors



a (3; 2)	b (2; -3)	a + b (5; -1)	|a| 3.61	|b| 3.61	|a + b| 5.1

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Scalar (dot) product	Skalyar ko‘paytma	Скалярное произведение
Angle between vectors	Vektorlar orasidagi burchak	Угол между векторами
Perpendicular vectors	Perpendikulyar vektorlar	Перпендикулярные векторы
Orthogonal	Ortogonal	Ортогональные
Projection	Proyeksiya	Проекция
Commutative	Kommutativ	Коммутативное
Distributive	Distributiv	Дистрибутивное
Direction cosines	Yo‘naltiruvchi kosinuslar	Направляющие косинусы
Work done	Bajarilgan ish	Работа силы

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$a \cdot b$ is:

- a vector
- a scalar
- an angle
- a matrix

$(1,2,3) \cdot (4,5,6)$ is:

- 32
- 12
- 21
- 90

$a \cdot b = 0$ means:

- parallel
- perpendicular
- equal
- opposite

$a \cdot a$ equals:

- 0
- $|a|$
- $|a|^2$
- $2a$

If $a \cdot b < 0$ the angle is:

- acute
- right
- obtuse
- zero

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(1,0,0) \cdot (0,1,0)$

ANSWER 0

2. $(1,2,3) \cdot (1,1,1)$

ANSWER 6

3. $(2,0,1) \cdot (3,4,5)$

ANSWER 11

4. $(1,1,1) \cdot (1,1,1)$

ANSWER 3

5. Are $(1,0,0)$ and $(0,0,1)$ perpendicular?

ANSWER yes

6. Angle between $(1,0,0)$ and $(1,0,0)$

ANSWER $0°$

7. Angle between $(1,0,0)$ and $(-1,0,0)$

ANSWER $180°$

Medium · the standard question

7 PROBLEMS

1. Angle between $(1,1,0)$ and $(1,0,1)$

   ANSWER $60°$

2. Angle between $(1,2,2)$ and $(2,-1,2)$

   ANSWER $\approx 63.6°$

3. Are $(2,1,-2)$ and $(1,2,2)$ perpendicular?

   ANSWER yes

4. Find t: $(t,2,3) \perp (4,t,-6)$

   ANSWER $t = 3$

5. Projection of $(3,4,0)$ on $(1,0,0)$

   ANSWER 3

6. Projection of $(1,2,2)$ on $(0,0,1)$

   ANSWER 2

7. $|a| = 5,\ |b| = 4,\ \theta = 60°$; find $a \cdot b$

   ANSWER 10

Hard · stretch and reason

7 PROBLEMS

1. Angle between the space diagonal and an edge of a unit cube

   ANSWER $\approx 54.7°$

2. Angle between two space diagonals of a unit cube

   ANSWER $\approx 70.5°$

3. Angle between a face diagonal and a space diagonal

   ANSWER $\approx 35.26°$

4. Find t so $(1, t, 2)$ makes $60°$ with $(1, 0, 0)$

   ANSWER $t = \pm\sqrt{11}$

5. Prove $|a + b|^2 = |a|^2 + 2a{\cdot}b + |b|^2$

   ANSWER Expand $(a+b){\cdot}(a+b)$

6. Deduce the cosine rule from the scalar product

   ANSWER $|a-b|^2 = |a|^2 + |b|^2 - 2|a||b|\cos\theta$

7. Prove the diagonals of a rhombus are perpendicular using vectors

   ANSWER $(a+b){\cdot}(a-b) = |a|^2 - |b|^2 = 0$

07 Homework

Homework — 5 tasks

Check the sign of $a \cdot b$ against your final angle every time.

1. Find $a \cdot b$ and the angle for $a = (2, -1, 3)$, $b = (1, 4, 1)$.
2. Find t so that $(3, t, -1)$ and $(2, 1, 5)$ are perpendicular.
3. Find the angle between the space diagonal of a cube and one of its edges.
4. Find the projection of $(4, 3, 0)$ on $(0, 1, 0)$.
5. Prove that $a \cdot a = |a|^2$ and use it to find $|(2, 3, 6)|$.

Transformations in space

Translation, reflection, rotation and enlargement — the plane transformations, each with one more coordinate.

LESSON 9 | 1 × 40 MIN | GEOMETRY 11, §5 | IGCSE E7.2

LESSON CLOCK

8' | 14' | 12' | 8' | 3'

Four transformations	8 min
Coordinate rules	14 min
Isometries	12 min
Composition	8 min
Homework	3 min

LEARNING OBJECTIVES

- Apply a translation, a reflection in a coordinate plane and an enlargement in space.
- Write the coordinate rule for each transformation.
- Recognise which transformations preserve length and which do not.
- Compose two transformations.

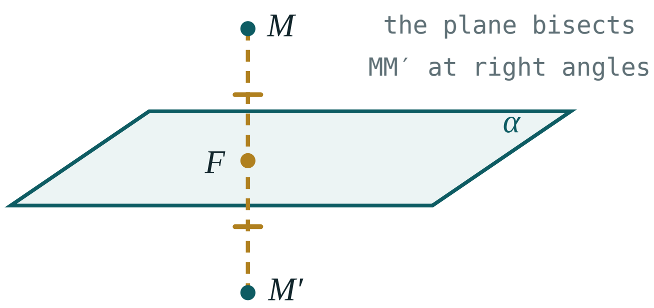
TEXTBOOK

National	pp. 47–52
Cambridge	Core & Extended, pp. 349–354
Workbook	Exercise 5.1

01 Explanation

Four transformations, four rules

TRANSFORMATION	RULE	PRESERVES LENGTH?
translation by (p, q, r)	$(x + p, y + q, z + r)$	yes
reflection in $z = 0$	$(x, y, -z)$	yes
reflection in $y = 0$	$(x, -y, z)$	yes
reflection in the z-axis	$(-x, -y, z)$	yes
reflection in the origin	$(-x, -y, -z)$	yes
rotation $90°$ about Oz	$(-y, x, z)$	yes
enlargement, centre O, factor k	(kx, ky, kz)	no — scaled by $\lvert k\rvert$



A point and its image in the plane z = 0. Only the third coordinate changes sign.

Isometries

DEFINITION

An **isometry** is a transformation preserving distance: $|A'B'| = |AB|$ for every pair. Translations, reflections and rotations are isometries; enlargements with $|k| \neq 1$ are not.

Isometries preserve everything that depends only on distance — angles, areas, volumes, congruence. An enlargement of factor k multiplies lengths by $|k|$, areas by k^2 and volumes by $|k|^3$.

A REFLECTION REVERSES ORIENTATION

A right-handed screw becomes left-handed in a mirror. That is why a reflection cannot be achieved by any rotation, however clever.

Composition

Applying two transformations in turn is itself a transformation. Two reflections in **parallel** planes give a translation of twice the gap; two reflections in **intersecting** planes give a rotation about their line of intersection, through twice the angle between them.

$$\text{reflect in } z = 0\text{, then in } z = 3 \Longrightarrow \text{translation by } (0, 0, 6)$$

Composition is not commutative in general: reflecting then translating is not the same as translating then reflecting.

02 Worked examples

EXAMPLE 1 **Find the image of $A(2, -3, 5)$ under reflection in the plane $y = 0$, then a translation by $(1, 1, 1)$.**

1. Reflection: $(2, 3, 5)$.
2. Translation: $(3, 4, 6)$.

ANSWER

$(3, 4, 6)$

EXAMPLE 2 **A solid of volume 40 cm³ is enlarged by factor 3. Find the new volume.**

1. Volumes scale by k^3.
2. 27×40

ANSWER

1080 cm³

EXAMPLE 3 **What single transformation results from reflecting in $x = 0$ then in $x = 4$?**

1. $(x,y,z) \to (-x,y,z)$

2. $\to (8 + x, y, z)$
   Reflection in $x = 4$ sends u to $8 - u$.

3. Net effect: add 8 to x.

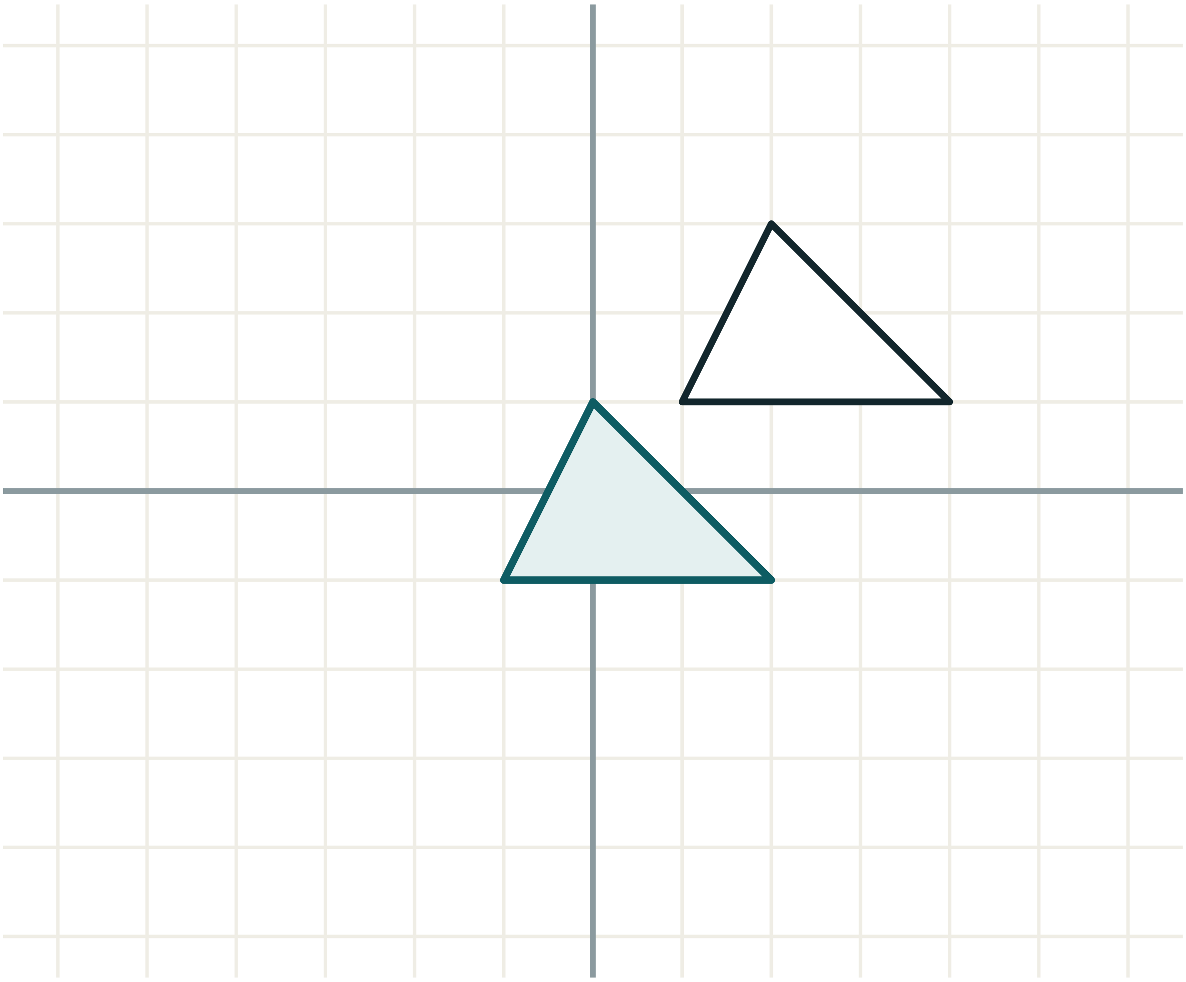
ANSWER

Translation by $(8, 0, 0)$

03 Interactive model

Reflect a shoe in a mirror to show that no rotation can reproduce a reflection.

Transforming a figure



transformation **translation** | lengths **unchanged** | angles **unchanged** | orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Transformation	Almashtirish	Преобразование
Translation	Parallel ko‘chirish	Параллельный перенос
Reflection in a plane	Tekislikka nisbatan akslantirish	Отражение относительно плоскости
Rotation about an axis	O‘q atrofida burish	Поворот вокруг оси
Enlargement (dilation)	Gomotetiya	Гомотетия
Centre of enlargement	Gomotetiya markazi	Центр гомотетии
Isometry	Izometriya	Изометрия
Invariant point	O‘zgarmas nuqta	Неподвижная точка
Image	Tasvir	Образ

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Reflection in $z = 0$ sends (x,y,z) to:

- $(-x,y,z)$
- $(x,-y,z)$
- $(x,y,-z)$
- $(-x,-y,-z)$

An enlargement of factor 2 multiplies volume by:

- 2
- 4
- 8
- 16

Which is not an isometry?

- translation
- reflection
- rotation
- enlargement of factor 3

Two reflections in parallel planes give:

- a rotation
- a translation
- an enlargement
- the identity

Reflection in the origin sends $(1,2,3)$ to:

- $(-1,2,3)$
- $(1,-2,-3)$
- $(-1,-2,-3)$
- $(3,2,1)$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Image of $(1,2,3)$ under reflection in $z = 0$

   ANSWER $(1,2,-3)$

2. Image under reflection in $x = 0$

   ANSWER $(-1,2,3)$

3. Image under reflection in the origin

   ANSWER $(-1,-2,-3)$

4. Image under translation $(2,2,2)$

   ANSWER $(3,4,5)$

5. Image under enlargement factor 2, centre O

   ANSWER $(2,4,6)$

6. Volume scale factor for $k = 3$

   ANSWER 27

7. Area scale factor for $k = 4$

   ANSWER 16

Medium · the standard question

7 PROBLEMS

1. Reflect $(2,-3,5)$ in $y = 0$, then translate by $(1,1,1)$

   ANSWER $(3,4,6)$

2. Rotate $(1,0,4)$ by $90°$ about Oz

   ANSWER $(0,1,4)$

3. Two reflections in $x = 0$ and $x = 4$

   ANSWER translation by $(8,0,0)$

4. A solid of volume 40 enlarged by 3 — new volume

   ANSWER 1080

5. Surface area 24 enlarged by 5 — new area

   ANSWER 600

6. Image of $(4, -2, 6)$ under enlargement factor $-\frac{1}{2}$

   ANSWER $(-2, 1, -3)$

7. Is reflection an isometry?

   ANSWER yes

Hard · stretch and reason

7 PROBLEMS

1. Find the reflection of $(1,2,3)$ in the plane $z = 5$

   ANSWER $(1,2,7)$

2. Find the reflection of $(3,1,2)$ in the plane $x = -1$

   ANSWER $(-5,1,2)$

3. Two reflections in planes at $30°$ — what results?

   ANSWER A rotation of $60°$

4. A cube of side 4 is enlarged so its volume becomes 1728. Find k

   ANSWER $k = 3$

5. Show that composing two translations gives a translation

   ANSWER Add the vectors

6. Why can no rotation reproduce a reflection?

   ANSWER Rotations preserve orientation; reflections reverse it

7. Find the centre of an enlargement mapping $(2,2,2)$ to $(6,6,6)$ with $k = 3$

   ANSWER $(0,0,0)$

07 Homework

Homework — 4 tasks

Write the coordinate rule beside every image you find.

1. Find the image of $(3, -1, 4)$ under reflection in each of the three coordinate planes.
2. Find the image of $(2, 5, -1)$ under enlargement of factor -2 about the origin.
3. A solid has volume 54 cm³ and surface area 90 cm². Find both after an enlargement of factor 2.
4. What single transformation is equivalent to reflecting in $y = 0$ and then in $y = 3$?

Symmetry in space

Central, axial and plane symmetry — the three kinds, their coordinate rules, and how to count them on a solid.

LESSON 10–11 | 2 × 40 MIN | GEOMETRY 11, §6 | IGCSE E4.6

LESSON CLOCK

10' | 20' | 24' | 20' | 14' | 2

Three kinds	10 min
Coordinate rules	20 min
Counting on solids	24 min
Symmetry as a shortcut	20 min
Practice	14 min
Homework	2 min

LEARNING OBJECTIVES

- Distinguish central, axial and plane symmetry.
- Write the coordinate rule for each and apply it.
- Count the symmetries of a given solid.
- Use symmetry to shorten a calculation or a proof.

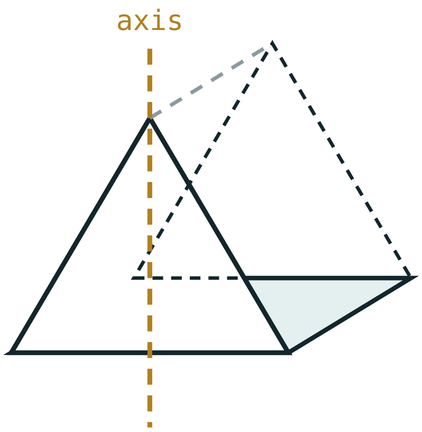
TEXTBOOK

National	pp. 53–64
Cambridge	Core & Extended, pp. 355–362
Workbook	Exercise 6.1

01 Explanation

Three kinds of symmetry

KIND	FIXED OBJECT	RULE (ABOUT THE ORIGIN / AXES)
central	a point	$(x, y, z) \to (-x, -y, -z)$
axial (order 2)	a line	$(x, y, z) \to (-x, -y, z)$ about Oz
plane (mirror)	a plane	$(x, y, z) \to (x, y, -z)$ in $z = 0$



a plane of symmetry cuts it into mirror halves

The same cuboid, seen through a plane of symmetry and turned about an axis.

CENTRAL SYMMETRY IS A HALF-TURN PLUS A MIRROR

In space, $(x,y,z) \to (-x,-y,-z)$ equals a $180°$ rotation about Oz followed by a reflection in $z = 0$. It reverses orientation, so it is not a rotation.

Counting on solids

SOLID	CENTRE?	PLANES	AXES
cube	yes	*9*	13
cuboid, edges different	yes	*3*	3
regular tetrahedron	no	*6*	7
square-based pyramid	no	*4*	1
regular hexagonal prism	yes	*7*	7
sphere	yes	infinitely many	infinitely many

A PYRAMID HAS NO CENTRE OF SYMMETRY

The apex would have to map to a point of the base, and there is nothing there. Any solid with a centre has every point paired with an opposite one — so a pyramid, a cone and a tetrahedron all fail.

Symmetry as a tool

Three uses that save real work:

- The **centre of mass** lies on every plane and axis of symmetry — so at their intersection.
- Two lengths or angles symmetric about a plane are **equal**, with no calculation.
- A section perpendicular to an axis of order n repeats n times, so one computation covers n cases.

Example. The centre of a cube with opposite vertices $(0,0,0)$ and $(6,6,6)$ is $(3,3,3)$ — by symmetry, in one line, with no distance formula.

02 Worked examples

EXAMPLE 1 **Find the image of $(2, -3, 4)$ under central symmetry about the origin, and about $(1, 1, 1)$.**

1. About O: negate all.
   $(-2, 3, -4)$
2. About C: $P' = 2C - P$.
3. $(2 - 2, 2 + 3, 2 - 4)$

ANSWER

$(-2, 3, -4)$ and $(0, 5, -2)$

EXAMPLE 2 **How many planes of symmetry has a regular hexagonal prism?**

1. The hexagon has 6 lines of symmetry.
   6 vertical planes.
2. One horizontal plane halfway up.

ANSWER

7

EXAMPLE 3 **Does a square-based pyramid have a centre of symmetry?**

1. A centre pairs every point with an opposite.
2. The apex would need a matching point below the base.
   There is none.

ANSWER

No

03 Interactive model

Hold a cuboid box and find each plane, then each axis, and count aloud.

Counting symmetry in space

Central symmetry sends $(1,2,3)$ to:

- $(-1,2,3)$
- $(1,-2,3)$
- $(-1,-2,-3)$
- $(3,2,1)$

A cube has how many planes of symmetry?

- 3
- 6
- 9
- 12

A square pyramid has a centre of symmetry:

- yes
- no
- sometimes
- only if regular

A regular hexagonal prism has how many planes?

- 6
- 7
- 12
- 13

The centre of mass lies on:

- no axis
- every plane of symmetry
- the base
- the apex

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Central symmetry	Markaziy simmetriya	Центральная симметрия
Axial symmetry	O‘qqa nisbatan simmetriya	Осевая симметрия
Plane symmetry	Tekislikka nisbatan simmetriya	Зеркальная симметрия
Centre of symmetry	Simmetriya markazi	Центр симметрии
Axis of symmetry	Simmetriya o‘qi	Ось симметрии
Plane of symmetry	Simmetriya tekisligi	Плоскость симметрии
Order of symmetry	Simmetriya tartibi	Порядок симметрии
Symmetric figure	Simmetrik shakl	Симметричная фигура
Symmetry group	Simmetriya guruhi	Группа симметрии

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Central symmetry about C sends P to:

- $C - P$
- $2C - P$
- $P - C$
- $-P$

Reflection in $z = 0$ changes:

- x
- y
- z
- all three

Which solid has no centre of symmetry?

- cube
- cuboid
- sphere
- tetrahedron

A cuboid with all edges different has how many axes?

- *0*
- *1*
- *3*
- *9*

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Central image of $(1,2,3)$

   ANSWER $(-1,-2,-3)$

2. Plane image of $(1,2,3)$ in $z = 0$

   ANSWER $(1,2,-3)$

3. Axial image about Oz of $(1,2,3)$

   ANSWER $(-1,-2,3)$

4. Planes of symmetry of a cube

   ANSWER 9

5. Planes of a $2 \times 3 \times 4$ cuboid

   ANSWER 3

6. Has a sphere a centre of symmetry?

   ANSWER yes

7. Has a cone a centre of symmetry?

   ANSWER no

Medium · the standard question

7 PROBLEMS

1. Central image of $(2,-3,4)$ about $(1,1,1)$

   ANSWER $(0,5,-2)$

2. Planes of a regular hexagonal prism

   ANSWER 7

3. Planes of a regular tetrahedron

   ANSWER 6

4. Axes of a cube

   ANSWER 13

5. Planes of a $3 \times 3 \times 8$ cuboid

   ANSWER 5

6. Centre of a cube with opposite vertices $(0,0,0)$, $(6,6,6)$

   ANSWER $(3,3,3)$

7. Rotational order of a regular pentagonal pyramid

   ANSWER 5

Hard · stretch and reason

7 PROBLEMS

1. Prove central symmetry equals a half-turn followed by a mirror

   ANSWER $(x,y,z) \to (-x,-y,z) \to (-x,-y,-z)$

2. Why has a regular tetrahedron 6 planes but no centre?

   ANSWER Each plane holds an edge and bisects the opposite; no vertex has an opposite

3. Find the centre of symmetry of the cuboid with opposite corners $(1,2,3)$ and $(7,10,15)$

   ANSWER $(4,6,9)$

4. A solid has exactly one plane of symmetry. Sketch one

   ANSWER e.g. an isosceles-triangle prism

5. How many symmetries (rotations only) has a cube?

   ANSWER 24

6. Show that a solid with a centre of symmetry has an even number of vertices

   ANSWER They pair up

7. Use symmetry to find the centroid of a regular tetrahedron of side a

   ANSWER At $\frac{1}{4}$ of the height above the base centre

07 Homework

Homework — 4 tasks

Sketch each solid and mark the symmetries on the sketch.

1. Find the image of $(3, -2, 5)$ under central symmetry about $(0,0,0)$ and about $(2, 1, 3)$.
2. Count the planes and axes of symmetry of a cube, a regular tetrahedron and a $4 \times 4 \times 9$ cuboid.
3. Explain why a pyramid can never have a centre of symmetry.
4. Use symmetry to find the centre of the cuboid with opposite corners $(-1, 0, 2)$ and $(5, 8, 10)$.

Symmetry in nature and in technology

Where the mathematics of the last two lessons actually appears — in crystals, in living things, and in the things engineers build.

LESSON 12 | 1 × 40 MIN | GEOMETRY 11, §7 | IGCSE E4.6

LESSON CLOCK

8' | 12' | 12' | 10' | 3'

Symmetry in living things	8 min
Crystals	12 min
Engineering and architecture	12 min
Uzbek ornament	10 min
Homework	3 min

LEARNING OBJECTIVES

- Identify the symmetry of a natural or manufactured object.
- Explain why a particular symmetry is useful in a design.
- Name the crystal systems by their symmetry.
- Present one object and its symmetry to the class.

TEXTBOOK

National	pp. 65–70
Cambridge	Core & Extended, pp. 363–366
Workbook	Project sheet S1

01 Explanation

Living things

ORGANISM	SYMMETRY	WHY IT HELPS
most animals	bilateral — one plane	a front and a direction of travel
starfish	radial, order 5	no preferred direction; food may come from anywhere
most flowers	radial, order 3–8	visible to a pollinator from any side
snowflake	radial, order 6	the hexagonal geometry of the water molecule
DNA	a helix — screw symmetry	compact storage of a long chain

A USEFUL OBSERVATION

Animals that **move** in a chosen direction are bilaterally symmetric; those that stay put, or drift, are radially symmetric. Motion selects a plane; stillness does not.

Crystals

A crystal is a lattice — one **unit cell** repeated by translation in three directions. The seven crystal systems are classified purely by their symmetry:

SYSTEM	CELL	EXAMPLE
cubic	$a = b = c$, all angles $90°$	salt, diamond, pyrite
tetragonal	$a = b \neq c$, all $90°$	zircon
orthorhombic	$a \neq b \neq c$, all $90°$	sulfur
hexagonal	a hexagonal prism cell	quartz, ice

A cubic cell carries all 9 planes and 13 axes of the cube — which is why salt cleaves into little cubes, and why a diamond can be cut along predictable planes.

Engineering, architecture, ornament

- **Wheels, gears, turbines** — radial symmetry of order n balances the rotating mass; an unbalanced wheel destroys its bearing.
- **Bridges and domes** — a plane of symmetry halves the load calculation and makes the structure predictable under wind from either side.
- **Aircraft** — bilateral symmetry keeps the centre of mass on the centre line.
- **Screw threads** — helical symmetry; the handedness matters, which is why a left-handed bolt is a real and separate object.

Uzbek ornament. The *girih* patterns of Samarkand and Bukhara are built from a small set of tiles with 5- and 10-fold rotational symmetry, repeated by translation. The tilework of the Registan is a mathematical object: a plane group, chosen and executed by hand centuries before the classification of such groups was written down.

THE PROJECT

Each learner brings one object — natural or made — photographs or sketches it, and states its symmetry precisely: how many planes, how many axes and of what order, and whether it has a centre. Two present next lesson.

02 Worked examples

EXAMPLE 1 State the symmetry of a starfish.

1. Five identical arms.
   Rotational order 5.
2. One mirror line through each arm.
   5 planes in three dimensions.
3. No centre — the top and bottom differ.

ANSWER

Radial, order 5, with 5 planes; no centre

EXAMPLE 2 Why must a turbine blade assembly be rotationally symmetric?

1. An asymmetric mass has a centre of mass off the axis.
2. That produces a rotating force on the bearings.
3. At high speed the force is destructive.

ANSWER

To keep the centre of mass on the rotation axis and avoid vibration

EXAMPLE 3 **A salt crystal is cubic. How many planes of symmetry has one grain?**

1. A cube has 3 face planes.
2. And 6 through opposite edges.

ANSWER

9

03 Interactive model

Bring a real object — a nut, a flower, a snowflake photograph — and count its symmetries aloud.

Symmetry around us

Most animals have:

- radial symmetry
- bilateral symmetry
- no symmetry
- a centre of symmetry

A snowflake has rotational order:

- 4
- 5
- 6
- 8

A cubic crystal has how many planes?

- 3
- 6
- 9
- 12

A screw thread has:

- plane symmetry
- helical symmetry
- central symmetry
- no symmetry

Girih patterns commonly show rotational order:

- 3 and 4
- 5 and 10
- 7
- 12 only

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Bilateral symmetry	Ikki tomonlama simmetriya	Двусторонняя симметрия
Radial symmetry	Radial simmetriya	Радиальная симметрия
Crystal lattice	Kristall panjara	Кристаллическая решётка
Unit cell	Elementar yacheyka	Элементарная ячейка
Helix	Spiral	Спираль
Tessellation	Parketlash	Паркет (замощение)
Balance (in design)	Muvozanat	Равновесие
Girih pattern	Girih naqshi	Гирих
Chirality	Xirallik	Хиральность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Radial symmetry suits organisms that:

- move fast forward
- stay in one place
- fly
- burrow

A unit cell is:

- a whole crystal
- the smallest repeating block
- a plane of symmetry
- an axis

A left-handed and a right-handed screw are:

- identical
- mirror images
- rotations of each other
- the same object

Symmetry in a bridge mainly helps with:

- appearance only
- load calculation and predictability
- cost of paint
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Symmetry of a starfish

   ANSWER radial, order 5

2. Symmetry of a human body (externally)

   ANSWER bilateral, one plane

3. Rotational order of a snowflake

   ANSWER 6

4. Symmetry of a wheel with 8 spokes

   ANSWER radial, order 8

5. Planes of a cubic salt crystal

   ANSWER 9

6. Symmetry of DNA

   ANSWER helical

7. Symmetry of a flower with 5 petals

   ANSWER radial, order 5

Medium · the standard question

7 PROBLEMS

1. Why are aircraft bilaterally symmetric?

   ANSWER To keep the centre of mass on the centre line

2. Why must a turbine be radially symmetric?

   ANSWER To avoid vibration from an off-axis mass

3. Name two cubic minerals

   ANSWER salt, pyrite (also diamond)

4. Which crystal system has $a = b \neq c$ with right angles?

   ANSWER tetragonal

5. What rotational orders appear in girih patterns?

   ANSWER 5 and 10

6. Why do most flowers have radial symmetry?

   ANSWER Pollinators approach from any direction

7. Give a manufactured object with exactly one plane of symmetry

   ANSWER e.g. a chair, a car exterior

Hard · stretch and reason

7 PROBLEMS

1. Explain why 5-fold symmetry cannot tile the plane periodically

   ANSWER Pentagons leave gaps; only orders 2, 3, 4, 6 tile

2. How do girih tilings achieve 5-fold local symmetry then?

   ANSWER By combining several tile shapes — a quasi-periodic arrangement

3. A crystal has 3 planes and a centre. Which system?

   ANSWER orthorhombic

4. Why is chirality important in chemistry?

   ANSWER Mirror-image molecules can behave completely differently in the body

5. Design a logo with rotational order 3 and no plane of symmetry

   ANSWER e.g. a three-bladed spiral

6. Count the symmetries of a standard hexagonal pencil

   ANSWER 7 planes, 7 axes

7. Why does a soap bubble take a spherical shape?

   ANSWER Maximum symmetry minimises surface area for a fixed volume

07 Homework

Homework — 4 tasks

Bring your object, or a clear photograph of it, to the next lesson.

1. Choose one object — natural or manufactured — and state its symmetry precisely: planes, axes with orders, and whether it has a centre.
2. Photograph or sketch a piece of Uzbek tilework and identify its rotational order.
3. Explain in three sentences why moving animals are bilaterally symmetric and stationary ones are radially symmetric.
4. Name the four crystal systems in the lesson with one example each.

Similarity of spatial figures

One scale factor, three consequences — lengths by k, areas by k^2, volumes by k^3.

LESSON 13 | 1 × 40 MIN | GEOMETRY 11, §8 | IGCSE E4.5

LESSON CLOCK

8' | 14' | 12' | 8' | 3'

What similarity means for solids	8 min
The three rules	14 min
Working backwards	12 min
Applications	8 min
Homework	3 min

LEARNING OBJECTIVES

- Decide whether two solids are similar.
- Apply the k, k^2, k^3 rules to lengths, areas and volumes.
- Work backwards from an area or volume ratio to the scale factor.
- Solve problems on models, maps and scaled containers.

TEXTBOOK

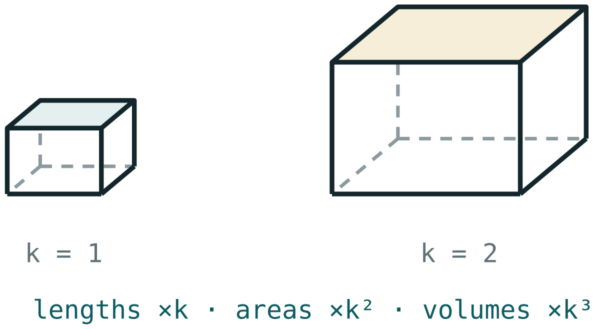
National	pp. 71–76
Cambridge	Core & Extended, pp. 367–372
Workbook	Exercise 8.1

01 Explanation

Similar solids

DEFINITION

Two solids are **similar** when one is an enlargement of the other: all corresponding lengths are in the same ratio k, and all corresponding angles are equal.



Two cuboids in the ratio 1 : 2. Every edge doubles; the surface quadruples; the volume is eight times.

All cubes are similar. All spheres are similar. Two cuboids are similar only if their three edge ratios all agree — a $2 \times 3 \times 4$ and a $4 \times 6 \times 8$ box are similar; a $2 \times 3 \times 4$ and a $4 \times 6 \times 9$ are not.

The three rules

$$\frac{L_2}{L_1} = k \cdot \frac{S_2}{S_1} = k^2 \cdot \frac{V_2}{V_1} = k^3$$

K	LENGTHS	AREAS	VOLUMES
2	$\times 2$	$\times 4$	$\times 8$
3	$\times 3$	$\times 9$	$\times 27$
$\frac{1}{2}$	$\times \frac{1}{2}$	$\times \frac{1}{4}$	$\times \frac{1}{8}$
1.5	$\times 1.5$	$\times 2.25$	$\times 3.375$

THE COMMONEST ERROR IN THE WHOLE TOPIC

Doubling the dimensions does **not** double the volume — it multiplies it by eight. A model car at $1 : 20$ needs $\frac{1}{8000}$ of the paint volume, not $\frac{1}{20}$.

Working backwards

Given a ratio of areas or volumes, recover k by taking the root:

$$k = \sqrt{\frac{S_2}{S_1}} \text{ or } k = \sqrt[3]{\frac{V_2}{V_1}}$$

Two similar cones have volumes 54 and 128 cm³. Then $k = \sqrt[3]{\frac{128}{54}} = \sqrt[3]{\frac{64}{27}} = \frac{4}{3}$, so the heights are in the ratio $3 : 4$ and the surface areas in the ratio $9 : 16$.

PRACTICAL CONSEQUENCES

A model at $1 : 50$ has $\frac{1}{2500}$ of the surface and $\frac{1}{125000}$ of the volume. This is why large animals have thick legs: mass grows as k^3 but bone cross-section only as k^2.

02 Worked examples

EXAMPLE 1 **Two similar cylinders have radii 3 cm and 5 cm. The smaller has volume 90 cm³. Find the larger.**

1. $k = \frac{5}{3}$
2. $k^3 = \frac{125}{27}$
3. $V = 90 \times \frac{125}{27}$
4. $= 416.7$

ANSWER

≈ 417 cm³

EXAMPLE 2 **Two similar solids have surface areas *48* and *108* cm². Find the ratio of their volumes.**

1. $k^2 = \dfrac{108}{48} = \dfrac{9}{4}$
2. $k = \dfrac{3}{2}$
3. $k^3 = \dfrac{27}{8}$

ANSWER

$27 : 8$

EXAMPLE 3 **A model aeroplane is built at *1 : 40*. The real wing area is *32* m². Find the model's wing area in cm².**

1. $k = \dfrac{1}{40}$, $k^2 = \dfrac{1}{1600}$
2. $32 \div 1600 = 0.02$ m².
3. $0.02 \times 10000 = 200$ cm².

ANSWER

200 cm²

03 Interactive model

Build two cubes of side 2 cm and 4 cm and count how many small ones fill the large one.

Scaling up and down

Doubling every length multiplies the volume by:

- 2
- 4
- 6
- 8

Doubling every length multiplies the surface area by:

- 2
- 4
- 8
- 16

Volumes in the ratio $27 : 64$ give lengths in:

- $27:64$
- $3:4$
- $9:16$
- $1:2$

Areas in the ratio $9:25$ give volumes in:

- $3:5$
- $9:25$
- $27:125$
- $81:625$

A $1:100$ model has what fraction of the volume?

- $\frac{1}{100}$
- $\frac{1}{10000}$
- $\frac{1}{1000000}$
- $\frac{1}{1000}$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Similar solids	O‘xshash jismlar	Подобные тела
Scale factor	O‘xshashlik koeffitsienti	Коэффициент подобия
Ratio of areas	Yuzalar nisbati	Отношение площадей
Ratio of volumes	Hajmlar nisbati	Отношение объёмов
Corresponding lengths	Mos uzunliklar	Соответственные длины
Model (scale model)	Model	Модель
Enlargement	Gomotetiya	Гомотетия
Linear dimension	Chiziqli o‘lcham	Линейный размер

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Similar solids have equal:

- volumes
- areas
- angles
- lengths

Areas scale by:

- k
- k^2
- k^3
- $2k$

Volumes scale by:

- k
- k^2
- k^3
- $3k$

All spheres are:

- congruent
- similar
- neither
- equal in volume

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $k = 2$; area factor

   ANSWER 4

2. $k = 2$; volume factor

   ANSWER 8

3. $k = 3$; volume factor

   ANSWER 27

4. $k = \frac{1}{2}$; volume factor

   ANSWER $\frac{1}{8}$

5. Volumes $8 : 27$; length ratio

   ANSWER $2 : 3$

6. Areas $4 : 9$; length ratio

   ANSWER $2 : 3$

7. Are all cubes similar?

   ANSWER yes

Medium · the standard question

7 PROBLEMS

1. Radii 3 and 5, smaller volume 90; larger volume

   ANSWER ≈ 417

2. Areas 48 and 108; volume ratio

   ANSWER $27 : 8$

3. Volumes 54 and 128; length ratio

   ANSWER $3 : 4$

4. $1 : 40$ model, real wing area 32 m²; model area

   ANSWER 200 cm²

5. A cone is cut halfway up. Volume ratio of small cone to whole

   ANSWER $1 : 8$

6. Same cut: ratio of small cone to frustum

   ANSWER $1 : 7$

7. Two similar tins: heights 10 and 15, smaller holds 400 ml; larger

   ANSWER 1350 ml

Hard · stretch and reason

7 PROBLEMS

1. A statue is $1{:}6$ scale and needs 0.5 litres of paint. The real one needs?

   ANSWER 18 litres

2. Two similar solids differ in volume by 117 cm³ with $k = \dfrac{4}{3}$. Find both volumes

   ANSWER 81 and 192

3. A cone of height 12 is cut at height 8. Volume ratio of the two parts

   ANSWER $8 : 19$

4. Why do large animals have proportionally thicker legs?

   ANSWER Mass grows as k^3, bone strength only as k^2

5. A $1{:}50$ model has surface area 80 cm². Find the real area in m²

   ANSWER 20 m²

6. Two similar cylinders have total surface areas 150π and 600π. Find the volume ratio

   ANSWER $1 : 8$

7. Show that if two solids have equal volumes and are similar they are congruent

   ANSWER $k^3 = 1 \Rightarrow k = 1$

07 Homework

Homework — 4 tasks

State k explicitly before using k^2 or k^3.

1. Two similar cones have heights 6 and 9 cm. The smaller has volume 80 cm³. Find the larger.
2. Two similar solids have surface areas 75 and 300 cm². Find the ratio of their volumes.
3. A model bridge is $1 : 200$. The real deck area is 4000 m². Find the model's deck area in cm².
4. Explain, using k^2 and k^3, why a doubled animal would break its own legs.

The vector equation of a line

Cambridge insert: a point and a direction give every point of a line — and intersection, parallelism and skewness all become arithmetic.

LESSON 14–16 | 3 × 40 MIN | GEOMETRY 11, §8 (EXTENSION) | P3 · 9.1–9.3

LESSON CLOCK

14' 22' 24' 26' 30' 19'

A point and a direction	14 min
From two points	22 min
Do two lines meet?	24 min
Parallel, intersecting or skew	26 min
The angle	30 min
Practice and homework	19 min

LEARNING OBJECTIVES

- Write the vector equation of a line from a point and a direction.
- Write it from two points, and convert to parametric form.
- Decide whether two lines meet, are parallel, or are skew.
- Find the angle between two lines and the point of intersection.

TEXTBOOK

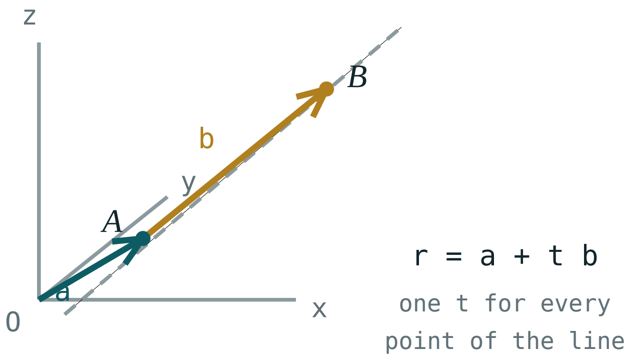
National	pp. 77–84
Cambridge	Pure Mathematics 2 & 3, pp. 190–208
Workbook	P3 Exercise 9A–9C

01 Explanation

A point and a direction

Start at a known point A with position vector a, and travel any multiple of a direction d:

$$r = a + t\,d, \quad t \in \mathbb{R}$$



A fixed point, a direction arrow, and the whole line traced out as t runs over the reals.

Each value of t gives one point; $t = 0$ gives A itself. In components, with $a = (1, 2, 3)$ and $d = (2, -1, 4)$:

$$x = 1 + 2t, y = 2 - t, z = 3 + 4t$$

THE EQUATION IS NOT UNIQUE

Any point of the line may play a, and any non-zero multiple of d may play the direction. $r = (3,1,7) + s(4,-2,8)$ is the **same line**. Never conclude two lines differ just because the equations look different.

From two points

Take one point as a and the vector between them as d:

$$r = a + t(b - a)$$

Through $A(1, 0, 2)$ and $B(4, 6, 8)$: $d = (3, 6, 6)$, or more simply $(1, 2, 2)$, so $r = (1, 0, 2) + t(1, 2, 2)$.

SIMPLIFY THE DIRECTION

Dividing d by a common factor changes nothing but makes every later calculation easier — and makes parallel lines obvious at a glance.

Do two lines meet?

THE PROCEDURE

Set the two position vectors equal, using **different** parameters t and s. That gives three equations in two unknowns. Solve two of them; substitute into the third. Consistent $\Longrightarrow$ they meet. Inconsistent $\Longrightarrow$ they do not.

$\ell_1 : r = (1,0,2) + t(1,2,2)$ and $\ell_2 : r = (2,5,3) + s(0,1,-1)$:

$$1 + t = 2 \Rightarrow t = 1;\ 2t = 5 + s \Rightarrow s = -3;\ check:\ 2 + 2t = 4,\ 3 - s = 6\ ✗$$

The third equation fails, so the lines do not meet. Their directions are not parallel either, so they are **skew**.

USE TWO DIFFERENT LETTERS

Writing t in both equations forces the two lines to reach the meeting point “at the same time”, which is an extra condition the geometry never imposed. It is the single most common error in this topic.

Classifying, and the angle

DIRECTIONS	SYSTEM	CONCLUSION
parallel	—	parallel (or identical, if a point is shared)
not parallel	consistent	they intersect — one point
not parallel	inconsistent	skew

The **angle** between two lines is the angle between their directions, from the scalar product — and it is defined even for skew lines, where the lines themselves never meet:

$$\cos\theta = \frac{|\mathrm{d}_1 \cdot \mathrm{d}_2|}{|\mathrm{d}_1||\mathrm{d}_2|}$$

The modulus in the numerator forces the acute angle, which is the conventional answer.

02 Worked examples

EXAMPLE 1 **Find the vector equation of the line through $A(2, -1, 3)$ and $B(4, 3, 7)$.**

1. $d = b - a = (2, 4, 4)$

   Simplify to $(1, 2, 2)$.

2. $r = (2, -1, 3) + t(1, 2, 2)$

3. Parametric: $x = 2 + t$, $y = -1 + 2t$, $z = 3 + 2t$.

ANSWER

$r = (2, -1, 3) + t(1, 2, 2)$

EXAMPLE 2 **Do $\ell_1 : r = (1,1,1) + t(2,1,3)$ and $\ell_2 : r = (3,2,4) + s(1,-1,2)$ meet?**

1. $1 + 2t = 3 + s$ and $1 + t = 2 - s$

2. Adding: $2 + 3t = 5 \Rightarrow t = 1$, then $s = 0$.

3. Third: $1 + 3t = 4$ and $4 + 2s = 4$.

   Both give 4 ✓

4. Point: $(3, 2, 4)$.

ANSWER

Yes, at $(3, 2, 4)$

EXAMPLE 3 **Find the acute angle between the lines with directions $(1, 2, 2)$ and $(2, -1, 2)$.**

1. $d_1 \cdot d_2 = 2 - 2 + 4 = 4$
2. $|d_1| = |d_2| = 3$
3. $\cos\theta = \dfrac{4}{9}$
4. $\theta \approx 63.6°$

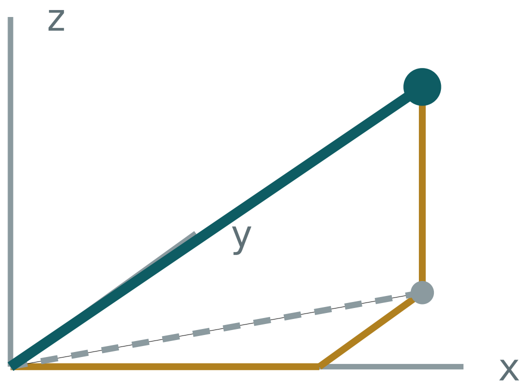
ANSWER

$\approx 63.6°$

03 Interactive model

Vary t and watch the point travel along the line; then add a second line and look for a crossing.

A line in space



x **3** | y **2** | z **2** | OM **4.12** | base diagonal **3.61**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Vector equation of a line	To‘g‘ri chiziq vektor tenglamasi	Векторное уравнение прямой
Direction vector	Yo‘naltiruvchi vektor	Направляющий вектор
Parameter	Parametr	Параметр
Parametric form	Parametrik shakl	Параметрический вид
Point of intersection	Kesishish nuqtasi	Точка пересечения
Skew lines	Ayqash chiziqlar	Скрещивающиеся прямые
Position vector	Radius-vektor	Радиус-вектор
Consistent system	Birgalikdagi sistema	Совместная система
Angle between lines	Chiziqlar orasidagi burchak	Угол между прямыми

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

In $r = a + td$, d is:

- a point
- the direction
- the length
- a parameter

The line through A and B is:

- $r = a + tb$
- $r = a + t(b - a)$
- $r = ab$
- $r = a - b$

When two lines are compared you must use:

- the same parameter
- two different parameters
- no parameter
- three parameters

Non-parallel directions with an inconsistent system means:

- intersecting
- parallel
- skew
- identical

The angle between lines uses:

- $|d_1 \cdot d_2|$
- $d_1 \cdot d_2$
- $|d_1| + |d_2|$
- the points

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Line through $(0,0,0)$ with direction $(1,1,1)$

   ANSWER $r = t(1,1,1)$

2. Line through $(1,2,3)$ with direction $(0,0,1)$

   ANSWER $r = (1,2,3) + t(0,0,1)$

3. Is $(2,4,6)$ on $r = t(1,2,3)$?

   ANSWER yes, $t = 2$

4. Is $(1,1,2)$ on $r = t(1,1,1)$?

   ANSWER no

5. Direction of the line through $(1,0,0)$ and $(3,2,4)$

   ANSWER $(2,2,4)$ or $(1,1,2)$

6. Are $(1,2,2)$ and $(2,4,4)$ parallel directions?

   ANSWER yes

7. Value of t giving the point a itself

   ANSWER $t = 0$

Medium · the standard question

7 PROBLEMS

1. Line through $A(2,-1,3)$ and $B(4,3,7)$

   ANSWER $r = (2,-1,3) + t(1,2,2)$

2. Parametric form of that line

   ANSWER $x = 2+t, y = -1+2t, z = 3+2t$

3. Do $r = (1,1,1)+t(2,1,3)$ and $r = (3,2,4)+s(1,-1,2)$ meet?

   ANSWER yes, at $(3,2,4)$

4. Angle between directions $(1,2,2)$ and $(2,-1,2)$

   ANSWER $\approx 63.6°$

5. Angle between $(1,0,0)$ and $(1,1,0)$

   ANSWER $45°$

6. Are $r = (0,0,0)+t(1,2,3)$ and $r = (1,2,3)+s(2,4,6)$ parallel?

   ANSWER yes — the same line

7. Where does $r = (1,2,3)+t(1,1,1)$ meet the plane $z = 0$?

   ANSWER $(-2,-1,0)$

Hard · stretch and reason

7 PROBLEMS

1. Show $r = (1,0,2)+t(1,2,2)$ and $r = (2,5,3)+s(0,1,-1)$ are skew

   ANSWER Directions not parallel; the system is inconsistent

2. Find where $r = (2,0,1)+t(1,1,2)$ meets $r = (4,2,5)+s(2,1,1)$

   ANSWER $(4,2,5)$

3. Find the angle between the lines in the previous question

   ANSWER $\approx 33.6°$

4. Find λ so $(3, \lambda, 6)$ lies on $r = (1,0,2)+t(1,2,2)$

   ANSWER $\lambda = 4$

5. Find the point of $r = t(1,2,2)$ closest to $(3,0,0)$

   ANSWER $(\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$

6. Show the lines through $(0,0,0)$,$(1,1,1)$ and $(1,0,0)$,$(0,1,1)$ intersect

   ANSWER At $(0.5, 0.5, 0.5)$

7. Write the equation of the x-axis in vector form

   ANSWER $r = t(1,0,0)$

07 Homework

Homework — 6 tasks

Use different letters for the two parameters, every time.

1. Find the vector equation of the line through $A(1, -2, 4)$ and $B(5, 2, 12)$.
2. Write that line in parametric form and find the point with $t = 3$.
3. Decide whether $r = (1,2,0) + t(1,1,1)$ and $r = (3,4,2) + s(2,1,3)$ meet, and if so where.
4. Find the acute angle between the directions $(2, 1, -2)$ and $(1, 2, 2)$.
5. Show that $r = (0,0,0) + t(1,1,0)$ and $r = (0,1,1) + s(1,0,0)$ are skew.

6. Find where $r = (2, 1, 5) + t(1, -1, 2)$ crosses the plane $y = 0$.

Control work, and the quarter review

Coordinates, vectors, the scalar product, symmetry and the vector line — one paper, then the map.

LESSON 17–18 | 2 × 40 MIN | GEOMETRY 11, NAZORAT ISHI 1 | P3 · CHAPTER 9 REVIEW

LESSON CLOCK

3 | 42' | 12' | 20' | 10' | 3

Instructions	3 min
The paper	42 min
Answers	12 min
Rewrite	20 min
Concept map	10 min
Targets	3 min

LEARNING OBJECTIVES

- Apply the whole quarter in one assessment.
- Choose between the coordinate and the vector method.
- Classify each lost mark and rewrite the solution.
- Build a concept map of vectors in space.

TEXTBOOK

National	pp. 85–88
Cambridge	Pure Mathematics 2 & 3, pp. 209–212
Workbook	Control paper G1

01 Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Distance and midpoint of $A(1,-2,4)$ and $B(5,2,16)$	4	L1–3
2	Centre and radius of $x^2+y^2+z^2-4x+2y-4=0$	3	L1–3
3	$a = (2,-1,2)$, $b = (1,2,2)$: find $\lvert a\rvert$, $a\cdot b$ and the angle	5	L7–8
4	Find t so that $(t,3,-1)$ and $(2,t,5)$ are perpendicular	3	L7–8
5	Are $A(1,0,2)$, $B(3,2,4)$, $C(6,5,7)$ collinear? Give $AB:BC$	4	L6
6	Vector equation of the line through $(2,-1,3)$ and $(4,3,7)$; is $(6,7,11)$ on it?	5	L14–16
7	Count the planes and axes of symmetry of a cube and of a $3\times3\times8$ cuboid	6	L10–11

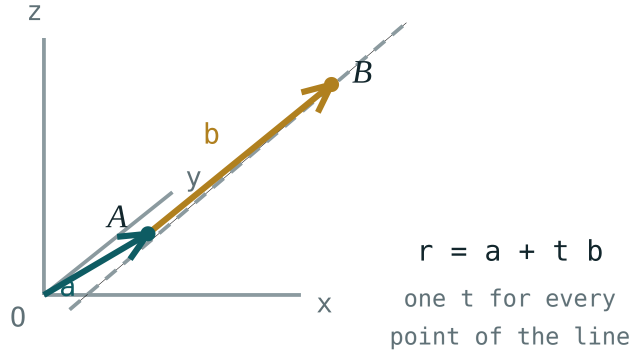
Q3 AND Q6 CARRY THE METHOD MARKS

Q3 gives 2 for the angle formula written correctly before any numbers go in. Q6 gives 2 for the direction vector and 3 for the test — a stated answer with no substitution scores one.

The concept map

Six boxes, links as sentences:

- **coordinates** → **distance** — “Pythagoras twice”
- **coordinates** → **position vectors** — “the same three numbers, read as a displacement”
- **vectors** → **collinear / coplanar** — “one scalar, or two”
- **vectors** → **scalar product** — “a number that carries the angle”
- **scalar product** → **perpendicularity** — “zero means a right angle”
- **point + direction** → **the line** — “$r = a + td$ generates every point”



The picture that ties the last three boxes together.

Looking forward

Quarter II is polyhedra, prisms and cylinders — surface area and volume. Every angle in it is found by the scalar product of this quarter, and every distance by the formula of Lesson 1–3. Neither is re-taught.

ONE HABIT TO CARRY FORWARD

Write the vector before the arithmetic. $AB = b - a$, then compute. Learners who go straight to numbers lose the sign of the direction and, with it, the angle.

02 Worked examples

EXAMPLE 1 **Model answer, Q3: $a = (2,-1,2)$, $b = (1,2,2)$.**

1. $|a| = \sqrt{4+1+4} = 3$
2. $|b| = \sqrt{1+4+4} = 3$
3. $a \cdot b = 2 - 2 + 4 = 4$
4. $\cos\theta = \dfrac{4}{9} \Rightarrow \theta \approx 63.6°$

ANSWER

$|a| = 3$, $a{\cdot}b = 4$, $\theta \approx 63.6°$

EXAMPLE 2 **Model answer, Q6: line through $(2,-1,3)$ and $(4,3,7)$; is $(6,7,11)$ on it?**

1. $d = (2,4,4)$, simplify to $(1,2,2)$.
2. $r = (2,-1,3) + t(1,2,2)$
3. $2 + t = 6 \Rightarrow t = 4$
4. Check: $-1 + 8 = 7$ ✓, $3 + 8 = 11$ ✓

ANSWER

Yes, at $t = 4$

EXAMPLE 3 **Model answer, Q2: centre and radius of $x^2+y^2+z^2-4x+2y-4=0$.**

1. $(x-2)^2 - 4$
2. $(y+1)^2 - 1$
3. $(x-2)^2 + (y+1)^2 + z^2 = 9$
4. Read off.

ANSWER

centre $(2,-1,0)$, radius 3

03 Interactive model

Work Q3 and Q6 on the board before the rewrite hour begins.

The quarter in ten questions

Distance $A(1,-2,4)$ to $B(5,2,16)$:

- 12
- 13
- 14
- 16

Midpoint of those points:

- $(3,0,10)$
- $(2,0,10)$
- $(3,0,8)$
- $(6,0,20)$

Radius of $(x-2)^2+(y+1)^2+z^2=9$:

- 9
- 3
- $\sqrt{3}$
- 81

$(2,-1,2) \cdot (1,2,2)$:

- 4
- 0
- 8
- -4

The angle between them:

- $30°$
- $45°$
- $\approx 63.6°$
- $90°$

$(t,3,-1) \perp (2,t,5)$ gives t:

- 1
- -1
- 5
- -5

$A(1,0,2)$, $B(3,2,4)$, $C(6,5,7)$ are:

- collinear
- not collinear
- coplanar only
- perpendicular

Direction of the line through $(2,-1,3)$ and $(4,3,7)$:

- $(1,2,2)$
- $(2,2,4)$
- $(1,1,1)$
- $(2,-1,3)$

Planes of symmetry of a $3\times3\times8$ cuboid:

- 3
- 4
- 5
- 9

Two non-parallel lines with an inconsistent system are:

- parallel
- intersecting
- skew
- identical

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Justification	Asoslash	Обоснование
Method choice	Usul tanlash	Выбор метода
Self-assessment	O‘z-o‘zini baholash	Самооценка
Revision	Takrorlash	Повторение
Target	Maqsad	Цель
Command word	Topshiriq so‘zi	Командное слово

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The most efficient test for a right angle in space:

- measure it
- the cosine rule
- $a \cdot b = 0$
- Pythagoras

Two lines are skew when:

- directions parallel
- the system is inconsistent and directions differ
- they meet
- they are equal

A vector equation of a line is unique:

- always
- never
- only in 2-D
- only through the origin

Quarter II begins with:

- the derivative
- polyhedra and volume
- probability
- trigonometry

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Distance $A(1,-2,4)$ to $B(5,2,16)$

ANSWER 13

2. Midpoint of those

ANSWER $(3,0,10)$

3. Radius of $x^2+y^2+z^2=49$

ANSWER 7

4. $(2,-1,2)\cdot(1,2,2)$

ANSWER 4

5. $|(2,-1,2)|$

ANSWER 3

6. Planes of a cube

ANSWER 9

7. Direction through $(2,-1,3)$ and $(4,3,7)$

ANSWER $(1,2,2)$

Medium · the standard question

7 PROBLEMS

1. Centre and radius of $x^2+y^2+z^2-4x+2y-4=0$

   ANSWER $(2,-1,0)$, 3

2. Angle between $(2,-1,2)$ and $(1,2,2)$

   ANSWER $\approx 63.6°$

3. Find t: $(t,3,-1) \perp (2,t,5)$

   ANSWER $t = 1$

4. Are $A(1,0,2)$, $B(3,2,4)$, $C(6,5,7)$ collinear? $AB:BC$

   ANSWER yes, $2:3$

5. Is $(6,7,11)$ on $r = (2,-1,3)+t(1,2,2)$?

   ANSWER yes, $t = 4$

6. Planes of a $3\times3\times8$ cuboid

   ANSWER 5

7. Axes of a cube

   ANSWER 13

Hard · stretch and reason

7 PROBLEMS

1. Find the point of $r = (2,-1,3)+t(1,2,2)$ closest to the origin

   ANSWER $\left(\frac{16}{9}, -\frac{13}{9}, \frac{19}{9}\right)$

2. Show two given lines are skew and find the angle between them

   ANSWER Inconsistent system; angle from $|d_1 \cdot d_2|$

3. Find the sphere through $(0,0,0)$, $(2,0,0)$, $(0,2,0)$, $(0,0,2)$

   ANSWER $(x-1)^2+(y-1)^2+(z-1)^2 = 3$

4. Prove the diagonals of a cube meet at its centre

   ANSWER All four have midpoint $\left(\frac{a}{2}, \frac{a}{2}, \frac{a}{2}\right)$

5. Angle between two space diagonals of a cube

   ANSWER $\approx 70.5°$

6. Find λ so $(1,2,\lambda)$ is coplanar with $(1,0,1)$ and $(0,1,1)$

   ANSWER $\lambda = 3$

7. Where does $r = (1,1,1)+t(2,-1,3)$ meet the plane $x = 5$?

   ANSWER $(5,-1,7)$

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter II.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.
3. Find the vector equation of the line through $(0, 1, -2)$ and $(4, 5, 6)$, and test whether $(6, 7, 10)$ is on it.

4. Write your target for Quarter II in one checkable sentence, and date it.